AIO · MDM · VALIDATION REPORT

20260929-145716

1h test · **5** device(s) · **14:57:18–15:57:18** (60 min) · clock offset **-0.1s** · mdm.dev.aioapp.com + mdm-stage.dev.aioapp.com

Firmware-client vitals as the MDM stored them, graded against adb on the same devices.

At a glance · all devices

11 of 12 fields match on every device. 1 needs attention.

WATCH Battery level — within ±0 % only 100% of the time live, 98% in chart history, on 0 of 5 devices

DEVICES	ACCURATE READINGS	DASHBOARD BEHIND DEVICE	TEST	READ ERRORS
5	**100.0%**	**21 s**	**60 min**	**9**
5 on live	2435 of 2435 checks, all fields	95% of the time under 3.2 min	sampled every 60 s	9 adb of 296 · 0 API of 305

FIELD	ALLOWED	RESULT	LIVE VALUE	CHART HISTORY	TYPICAL ERROR	WORST ERROR	DEVICES OFF
Battery level `battery_pct`	±0 %	WATCH	100.0% (155/155)	97.6% (83/85)	±0%	±0%	none
Charging state `charging`	exact	PASS	100.0% (155/155)	100.0% (85/85)	—	—	none
Charger type `charger_type`	exact	PASS	100.0% (155/155)	—	—	—	none
Battery present `battery_present`	exact	PASS	100.0% (155/155)	—	—	—	none
Battery voltage `charger_voltage_mv`	±75 mV	PASS	100.0% (155/155)	—	±13 mV	±38 mV	none
Battery temperature `battery_temp_c`	±1 °C	PASS	100.0% (155/155)	100.0% (85/85)	±0.1 °C	±0.6 °C	none
Wireless charging `wlc_charging`	exact	PASS	100.0% (155/155)	—	—	—	none
Wireless pad status `wlc_status`	exact (1 s watcher, 120 s cache)	PASS	100.0% (282/282)	100.0% (240/240)	—	—	none
RAM total `ram_total_mb`	±1 MB	PASS	100.0% (262/262)	100.0% (191/191)	±0 MB	±0 MB	none
RAM available `ram_available_mb`	±5 % of total	PASS	100.0% (262/262)	—	±9 MB	±63 MB	none
RAM used `ram_used_mb`	±5 % of total	PASS	100.0% (262/262)	100.0% (191/191)	±9 MB	±64 MB	none
Free storage `storage_free_gb`	±0.1 GB	PASS	100.0% (282/282)	100.0% (240/240)	±0 GB	±0.1 GB	none

Live value: the MDM's current value checked against adb when the device built it. Chart history: the stored rows behind the dashboard charts. Pass = 100%, Watch = 95% or more, Fail = under 95%. Errors are measured against the closest adb reading in the field's cache window.

Tolerances

Change an allowance and the summary above and sections 1 and 3 below re-grade on the spot. The miss lists below each table stay at the default rules. The link keeps your values, so it can be shared.

`battery_pct` ± 0 % · `charger_voltage_mv` ± 75 mV · `battery_temp_c` ± 1.0 °C · `ram_total_mb` ± 1 MB · `ram_available_mb` ± 5 % of total · `ram_used_mb` ± 5 % of total · `storage_free_gb` ± 0.1 GB — Reset to defaults

Details

Per device and per field. The summary above is built from these.

Devices

SERIAL	MDM	ADB TARGET	ADB SAMPLES	ADB ERRORS	API READS	API ERRORS	DISTINCT FRAMES	FRAME AGE P50 / P95 / MAX (CHARGING)	(ON BATTERY)
AT070AABU00064	live	10.32.0.134:5555	59	2	61	0	61	—	15s / 52s / 2.3m
AT070AABU00231	live	10.32.3.29:5555	59	2	61	0	54	—	31s / 6.2m / 9.2m
AT070AABU00280	live	10.32.0.13:5555	59	2	61	0	56	—	14s / 4.2m / 7.2m
AT070AABU00833	live	10.32.3.30:5555	60	1	61	0	54	—	17s / 4.2m / 7.2m
AT070AABU00875	live	10.32.3.31:5555	59	2	61	0	60	—	25s / 61s / 72s

1 · Accuracy at frame time

Each distinct frame, placed on the adb timeline by its `uptime_seconds`, checked against the adb snapshots within the field's cache window. Anything under 100% here is a real mismatch. Under each rate: the observed deviation (MDM − adb, against the best-matching snapshot in the window): its range, its median size and its largest size, to compare with the Allowed column.

FIELD	ALLOWED	AT070AABU00064	AT070AABU00231	AT070AABU00280	AT070AABU00833	AT070AABU00875
battery_pct	±0 %	100.0% (36/36) observed ±0 % (all exact)	100.0% (24/24) observed ±0 % (all exact)	100.0% (35/35) observed ±0 % (all exact)	100.0% (34/34) observed ±0 % (all exact)	100.0% (26/26) observed ±0 % (all exact)
charging	exact	100.0% (36/36)	100.0% (24/24)	100.0% (35/35)	100.0% (34/34)	100.0% (26/26)
charger_type	exact	100.0% (36/36)	100.0% (24/24)	100.0% (35/35)	100.0% (34/34)	100.0% (26/26)
battery_present	exact	100.0% (36/36)	100.0% (24/24)	100.0% (35/35)	100.0% (34/34)	100.0% (26/26)
charger_voltage_mv	±75 mV	100.0% (36/36) observed −23…+38 mV · median ±16 mV · max ±38 mV	100.0% (24/24) observed −30…+21 mV · median ±14 mV · max ±30 mV	100.0% (35/35) observed −19…+31 mV · median ±15 mV · max ±31 mV	100.0% (34/34) observed −24…+37 mV · median ±9 mV · max ±37 mV	100.0% (26/26) observed −25…+27 mV · median ±12 mV · max ±27 mV
battery_temp_c	±1 °C	100.0% (36/36) observed −0.4…+0.4 °C · median ±0.1 °C · max ±0.4 °C	100.0% (24/24) observed −0.2…+0.3 °C · median ±0 °C · max ±0.3 °C	100.0% (35/35) observed −0.2…+0.2 °C · median ±0.1 °C · max ±0.2 °C	100.0% (34/34) observed −0.1…+0.4 °C · median ±0 °C · max ±0.4 °C	100.0% (26/26) observed −0.6…+0.4 °C · median ±0 °C · max ±0.6 °C
wlc_charging	exact	100.0% (36/36)	100.0% (24/24)	100.0% (35/35)	100.0% (34/34)	100.0% (26/26)
wlc_status	exact (1 s watcher, 120 s cache)	100.0% (60/60)	100.0% (53/53)	100.0% (55/55)	100.0% (54/54)	100.0% (60/60)
ram_total_mb	±1 MB	100.0% (56/56) observed ±0 MB (all exact)	100.0% (45/45) observed ±0 MB (all exact)	100.0% (51/51) observed ±0 MB (all exact)	100.0% (53/53) observed ±0 MB (all exact)	100.0% (57/57) observed ±0 MB (all exact)
ram_available_mb	±5 % of total	100.0% (56/56) observed −38…+63 MB · median ±11 MB · max ±63 MB (1.7% of total)	100.0% (45/45) observed −43…+44 MB · median ±8 MB · max ±44 MB (1.2% of total)	100.0% (51/51) observed −31…+24 MB · median ±9 MB · max ±31 MB (0.9% of total)	100.0% (53/53) observed −27…+46 MB · median ±7 MB · max ±46 MB (1.3% of total)	100.0% (57/57) observed −40…+43 MB · median ±11 MB · max ±43 MB (1.2% of total)
ram_used_mb	±5 % of total	100.0% (56/56) observed −64…+38 MB · median ±11 MB · max ±64 MB (1.8% of total)	100.0% (45/45) observed −44…+42 MB · median ±8 MB · max ±44 MB (1.2% of total)	100.0% (51/51) observed −25…+30 MB · median ±9 MB · max ±30 MB (0.8% of total)	100.0% (53/53) observed −47…+26 MB · median ±7 MB · max ±47 MB (1.3% of total)	100.0% (57/57) observed −43…+40 MB · median ±11 MB · max ±43 MB (1.2% of total)
storage_free_gb	±0.1 GB	100.0% (60/60) observed ±0 GB (all exact)	100.0% (53/53) observed ±0 GB (all exact)	100.0% (55/55) observed ±0 GB (all exact)	100.0% (54/54) observed +0…+0.1 GB · median ±0 GB · max ±0.1 GB	100.0% (60/60) observed +0…+0.1 GB · median ±0 GB · max ±0.1 GB

2 · Freshness: how long the dashboard disagreed with the device

% of observed time the server's current value was outside tolerance of the device's current value, and the longest such stretch. Expected on battery: the client stretches its poll to ≥5 min while the WebSocket is healthy.

FIELD	AT070AABU00064	AT070AABU00231	AT070AABU00280	AT070AABU00833	AT070AABU00875
battery_pct	0.0% · longest 0s at 14:57:23 (battery: MDM 36% vs 35%)	0.0%	1.9% · longest 0s at 15:37:22 (battery: MDM 34% vs 33%)	0.0%	0.0%
charging	0.0%	0.0%	0.0%	0.0%	0.0%
charger_type	0.0%	0.0%	0.0%	0.0%	0.0%
battery_present	0.0%	0.0%	0.0%	0.0%	0.0%
charger_voltage_mv	0.0%	0.0%	0.0%	0.0%	0.0%
battery_temp_c	0.0%	0.0%	0.0%	0.0%	0.0%
wlc_charging	0.0%	0.0%	0.0%	0.0%	0.0%
wlc_status	0.0%	0.0%	0.0%	0.0%	0.0%
ram_total_mb	0.0%	0.0%	0.0%	0.0%	0.0%
ram_available_mb	0.0%	0.0%	0.0%	0.0%	0.0%
ram_used_mb	0.0%	0.0%	0.0%	0.0%	0.0%
storage_free_gb	0.0%	0.0%	0.0%	0.0%	0.0%

3 · History rows (what the charts and CSV read)

The stored samples returned by `GET /api/v1/devices/{serial}` inside the collection window, graded the same way. Row gaps below.

FIELD	ALLOWED	AT070AABU00064	AT070AABU00231	AT070AABU00280	AT070AABU00833	AT070AABU00875
battery_pct	±0 %	100.0% (18/18) observed ±0 % (all exact)	94.4% (17/18) observed −2…+0 % · median ±0 % · max ±2 %	92.3% (12/13) observed −1…+0 % · median ±0 % · max ±1 %	100.0% (15/15) observed ±0 % (all exact)	100.0% (21/21) observed ±0 % (all exact)
charging	exact	100.0% (18/18)	100.0% (18/18)	100.0% (13/13)	100.0% (15/15)	100.0% (21/21)
charger_type	exact	—	—	—	—	—
battery_present	exact	—	—	—	—	—
charger_voltage_mv	±75 mV	—	—	—	—	—
battery_temp_c	±1 °C	100.0% (18/18) observed −0.5…+0.3 °C · median ±0 °C · max ±0.5 °C	100.0% (18/18) observed −0.5…+0.3 °C · median ±0.1 °C · max ±0.5 °C	100.0% (13/13) observed −0.1…+0.4 °C · median ±0 °C · max ±0.4 °C	100.0% (15/15) observed −0.1…+0.3 °C · median ±0 °C · max ±0.3 °C	100.0% (21/21) observed −0.4…+0.5 °C · median ±0.1 °C · max ±0.5 °C
wlc_charging	exact	—	—	—	—	—
wlc_status	exact (1 s watcher, 120 s cache)	100.0% (53/53)	100.0% (44/44)	100.0% (47/47)	100.0% (46/46)	100.0% (50/50)
ram_total_mb	±1 MB	100.0% (39/39) observed ±0 MB (all exact)	100.0% (40/40) observed ±0 MB (all exact)	100.0% (28/28) observed ±0 MB (all exact)	100.0% (39/39) observed ±0 MB (all exact)	100.0% (45/45) observed ±0 MB (all exact)
ram_available_mb	±5 % of total	—	—	—	—	—
ram_used_mb	±5 % of total	100.0% (39/39) observed −64…+47 MB · median ±8 MB · max ±64 MB (1.8% of total)	100.0% (40/40) observed −44…+31 MB · median ±7 MB · max ±44 MB (1.2% of total)	100.0% (28/28) observed −38…+23 MB · median ±9.5 MB · max ±38 MB (1.0% of total)	100.0% (39/39) observed −47…+26 MB · median ±8 MB · max ±47 MB (1.3% of total)	100.0% (45/45) observed −43…+24 MB · median ±9 MB · max ±43 MB (1.2% of total)
storage_free_gb	±0.1 GB	100.0% (53/53) observed −0.1…+0 GB · median ±0 GB · max ±0.1 GB	100.0% (44/44) observed ±0 GB (all exact)	100.0% (47/47) observed ±0 GB (all exact)	100.0% (46/46) observed +0…+0.1 GB · median ±0 GB · max ±0.1 GB	100.0% (50/50) observed +0…+0.1 GB · median ±0 GB · max ±0.1 GB

AT070AABU00231 misses

(first few per field)

TIME	FIELD	MDM	ADB (NEAREST)
15:08:31	battery_pct	58%	60%

AT070AABU00280 misses

(first few per field)

TIME	FIELD	MDM	ADB (NEAREST)
15:19:30	battery_pct	35%	36%

SERIAL	ROWS	GAP P50	GAP P95	GAP MAX
AT070AABU00064	53	70s	110s	118s
AT070AABU00231	44	70s	110s	2.2m
AT070AABU00280	47	70s	99s	2.0m
AT070AABU00833	46	70s	108s	2.0m
AT070AABU00875	50	70s	100s	2.0m

Known client behaviour (not bugs in this test)

- `battery_pct` is the level the firmware broadcasts: `BatteryService` remaps 90–98% to 90–100% for display, so `dumpsys battery` (raw) can read 99 while the status bar and the MDM say 100. adb values are remapped the same way before comparing.
- `charger_voltage_mv` is `EXTRA_VOLTAGE` — the battery voltage, not the charger's.
- `battery_temp_c` is healthd's reading as-is (client 1.4.8+), graded against adb's; clients 1.4.6–1.4.7 reported a 3-minute average and will miss on every jump. RAM is cached 30 s, storage 120 s.
- On battery with a healthy WebSocket the client polls every ≥5 min; charging edges and WLC changes are pushed immediately.
- The server stores no device timestamp; this report times frames by `uptime_seconds` vs `/proc/uptime`.